

Qatar-2b

R-1679

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_190513 · created 2026-08-02T11:26:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458616.73517 +/- 0.00081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.222 +/- 0.065
Transit depth (Rp/Rs)^2	4.94 +/- 2.91 [%]
Orbital Inclination (inc)	89.89 +/- 1.72
Airmass coefficient 1 (a1)	0.5 +/- 0.29
Airmass coefficient 2 (a2)	0.41 +/- 0.88
Scatter in the residuals of the lightcurve fit is	54.36 %
Best Comparison Star	None
Optimal Aperture	10.29
Optimal Annulus	26.43
Transit Duration (day)	0.08 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.222 $\pm$ 0.065 vs 0.16327 (deviation 0.72 σ)
Duration fit vs published	0.08 d vs 0.0767 d (deviation 0.57 σ)
Mid-time O-C	1.8 min
Depth SNR	2.7
Residual scatter	54.36 %

Light curve

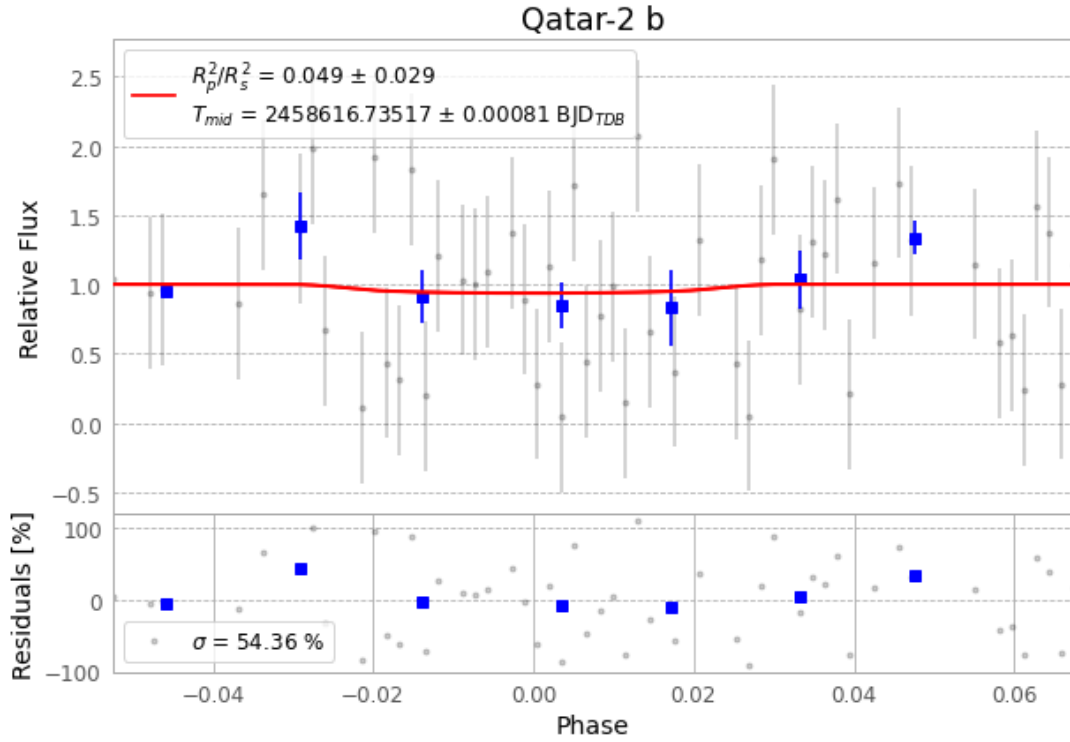


Figure 1 — FinalLightCurve_Qatar-2 b_2019-05-12.png (SHA-256 25d9303012782d0b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [413, 331], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52c3c9e9e62cf4f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), Qatar-2190513034812.FITS → Qatar-2190513073913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-190513.log (fc50aeee9f9e...), FinalLightCurve_Qatar-2 b_2019-05-12.png (25d930301278...), FinalLightCurve_Qatar-2 b_2019-05-12.csv (8107b5100c8b...)

Compute resources

Wall time 220.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 11:26Z.